PetLink

Smart Pet Health Monitoring for Busy Cat Owners



Session 3 — Group 25

The Problem: Busy Owners Cannot Check Their Pets in Time



Target Users: The Busy Pet Parents

Hardworking students and office workers who spend long hours away from home. Their daily schedules often leave them with limited time to care for their furry friends.

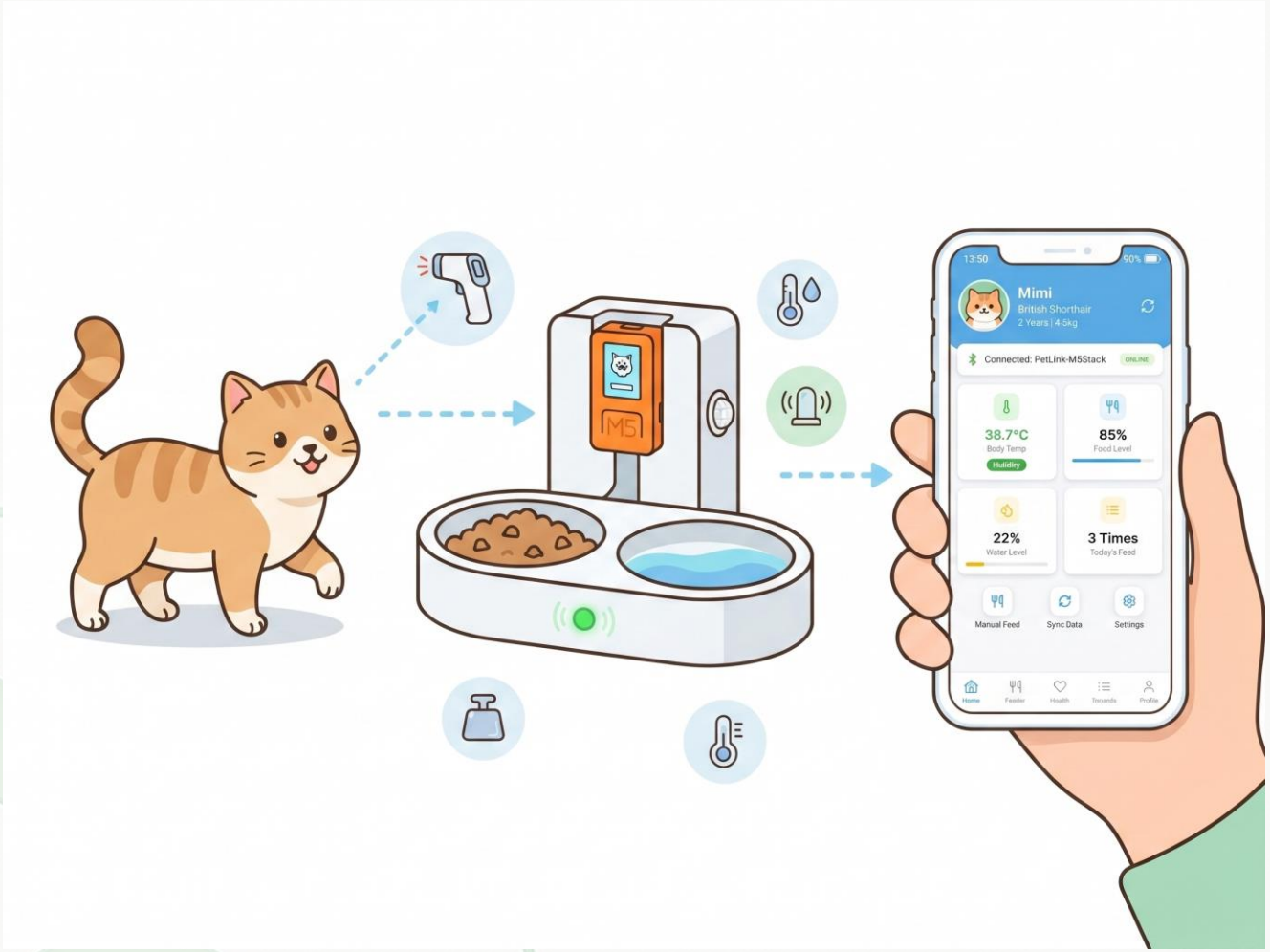
Core Pain Points

- Unable to confirm if their cat has enough food and water in real-time.
- No way to monitor their pet's health status or detect anomalies promptly.

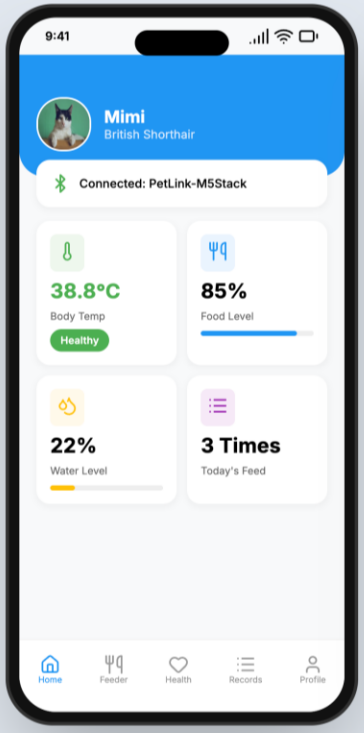
"I can see the numbers, but I'm not sure what I should do next."

What We Built: PetLink

PetLink, an all-in-one smart health monitoring system for busy cat owners. It automatically tracks your cat's body temperature, food and water levels, and daily eating habits, and sends real-time alerts to your phone. With PetLink, you can care for your pet even when you're away from home.

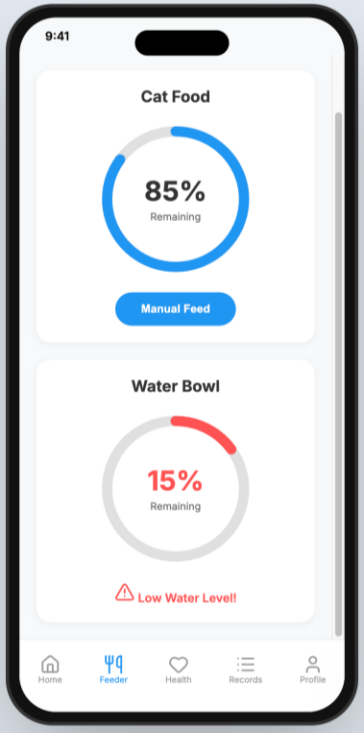


Live Demo: Software



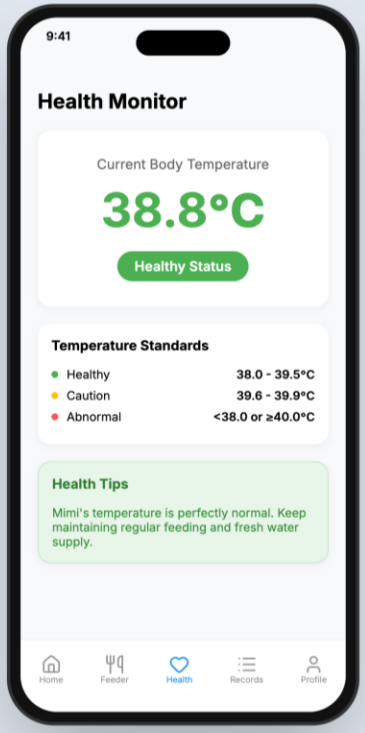
Home Dashboard

Overview of pet health status
Keep track of your pet's daily updates



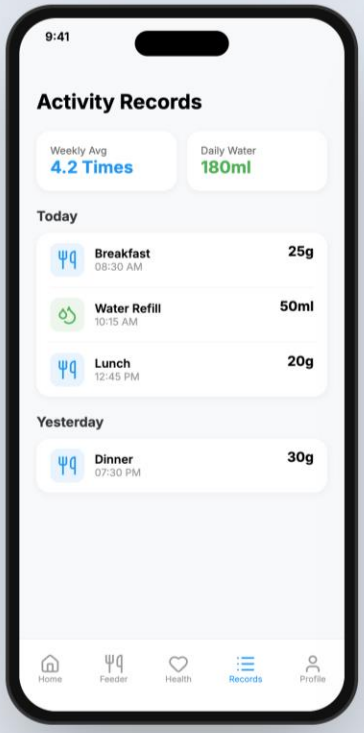
Feeder & Storage

Monitor food & water levels remotely
One-tap feeding, care always online



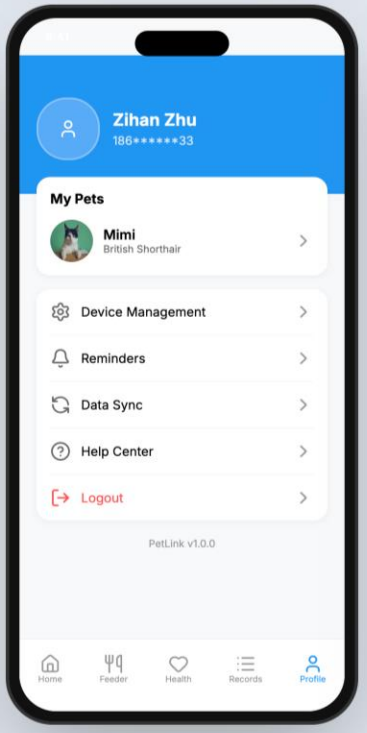
Health Monitoring

Track detailed health metrics like body temperature
Scientific insights for worry-free care



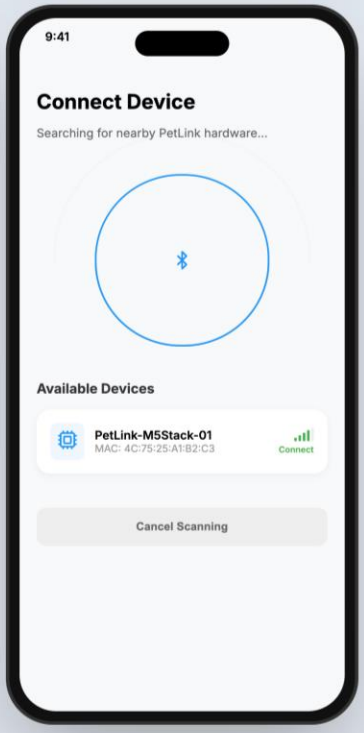
Activity Records

View complete feeding & drinking history
Reliable records for healthy habits



User Profile

Manage pets and device settings
Personalized experience tailored for you



Device Connection

Seamlessly connect to PetLink hardware
One-click pairing, ready in seconds

Live Demo: Hardware

4 Main Functions + 2 Additional Features

01 Body Temperature



Equipped with the NCIR infrared sensor, the main controller is able to monitor the body temperature of the cat when it approaches. Then the results will be transmitted to the PetLink app, allowing the owner to check the pet's health condition in real time and receive alerts for abnormal temperatures.



02 Food Level



Equipped with a high-precision Weight sensor, it accurately calculates and displays the remaining weight of dry cat food. The data is synced to the PetLink app, which sends timely refill reminders to ensure your pet never runs out of food.

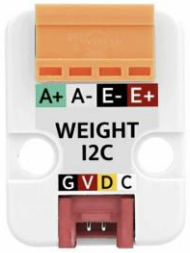


ENV.III Sensor
Detects humidity inside the feeder



RGB LED
Display dynamic lights

03 Water Level



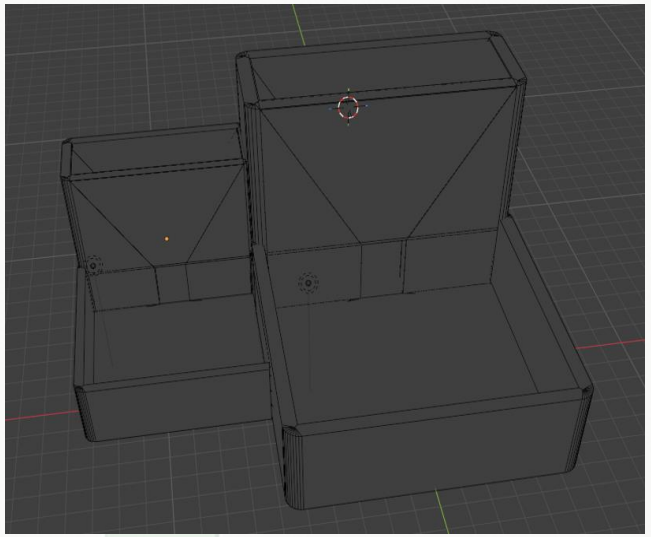
Similar with the food level detector, the weight sensor will also continuously monitor the remaining water level in the bowl and send data to the app, helping prevent dehydration.

04 Feeding Records

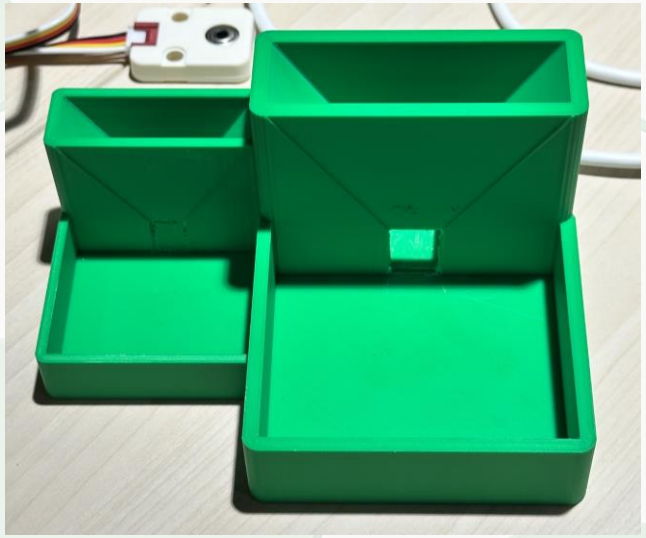


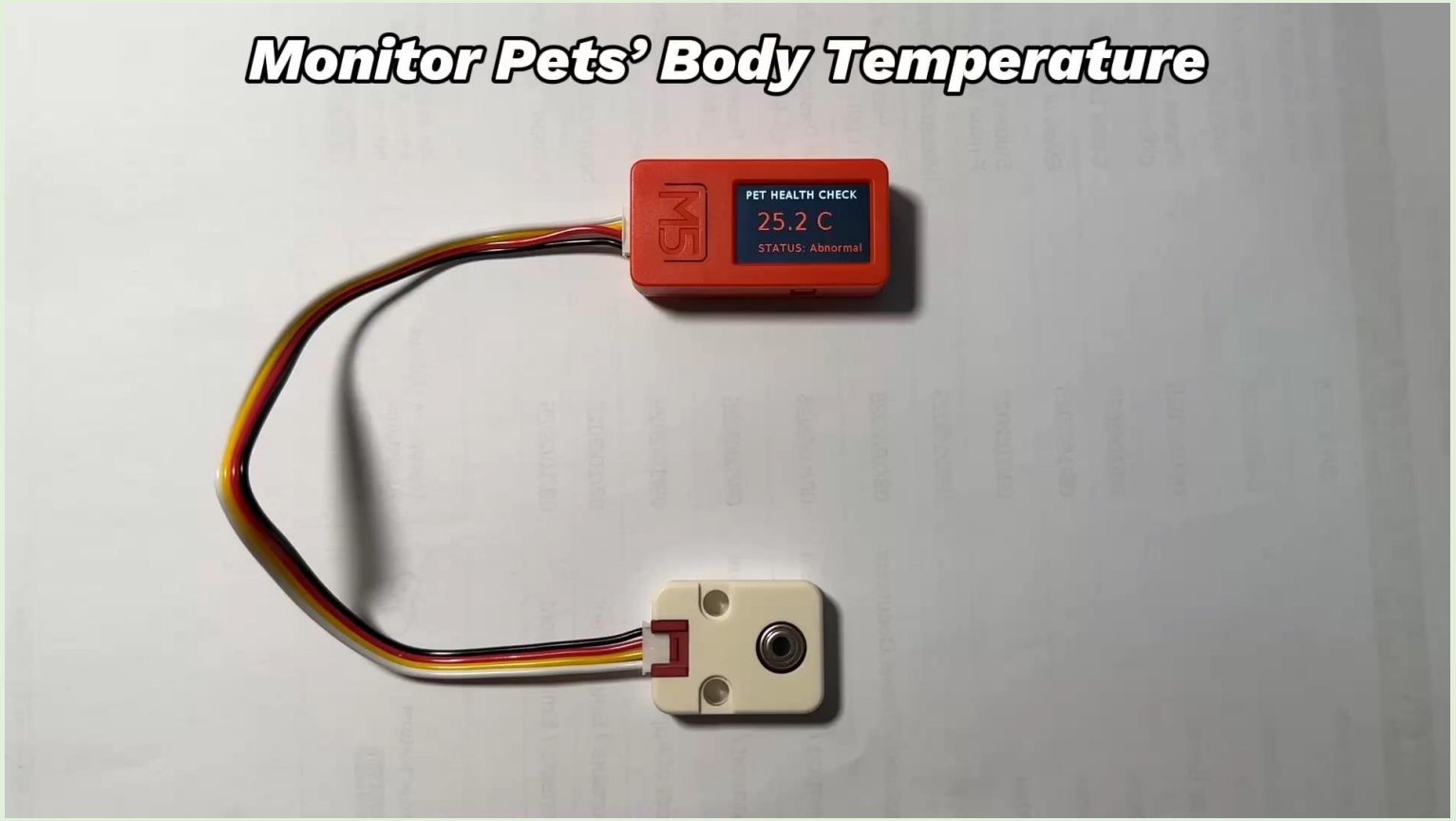
Based on the PIR Motion sensor, the time and amount of every meal and drink will be automatically recorded. These records are compiled into intuitive data charts in the app, helping owners understand and cultivate healthy eating habits for their pets.

Live Demo: Hardware (Modeling & Bottleneck)



3D Printing





We discovered a low-level compatibility bug in the official UIFlow2 firmware that affects the Pb.HUB expansion board. When multiple GPIO sensors are connected at the same time, the Pb.HUB's protocol conversion causes I2C bus congestion, leading to system crashes.

Instead of compromising our demo, we validated every core sensor individually and recorded a short video showing each function working perfectly.

Validation: From Device-Centred to Pet-Centred

BEFORE: Device-Centred

Data without Actionable Insight

❌ **Information overload with a lack of guidance:** The interface is cluttered with raw device parameters, lacking logical categorization and visual hierarchy. This makes it difficult for users to quickly focus on key information.

❌ **Passive reception state:** Users can only "see" cold data; they cannot intuitively understand the pet's condition behind the numbers, nor do they know what actions to take based on the data.

AFTER: Pet-Centred

Actionable & Intuitive Experience

✅ **Intuitive Home Dashboard:** Core metrics like pet health status and activity levels have been reorganized for instant comprehension.

✅ **Visual Status Cards:** Prominent status cards combining colors and icons provide intuitive cues about the pet's current condition, lowering the cognitive load.

✅ **Task-Oriented Navigation Optimization:** The bottom navigation structure has been restructured to prioritize high-frequency entry points like feeding and health reports, shortening the user journey.

What Users Still Told Us

It feels confusing that "22% water remaining" is labeled as "Normal".

Interpretation: Users find the current device status feedback unintuitive. They require a more explicit, tiered status indicator system to clearly convey the situation and alleviate anxiety.

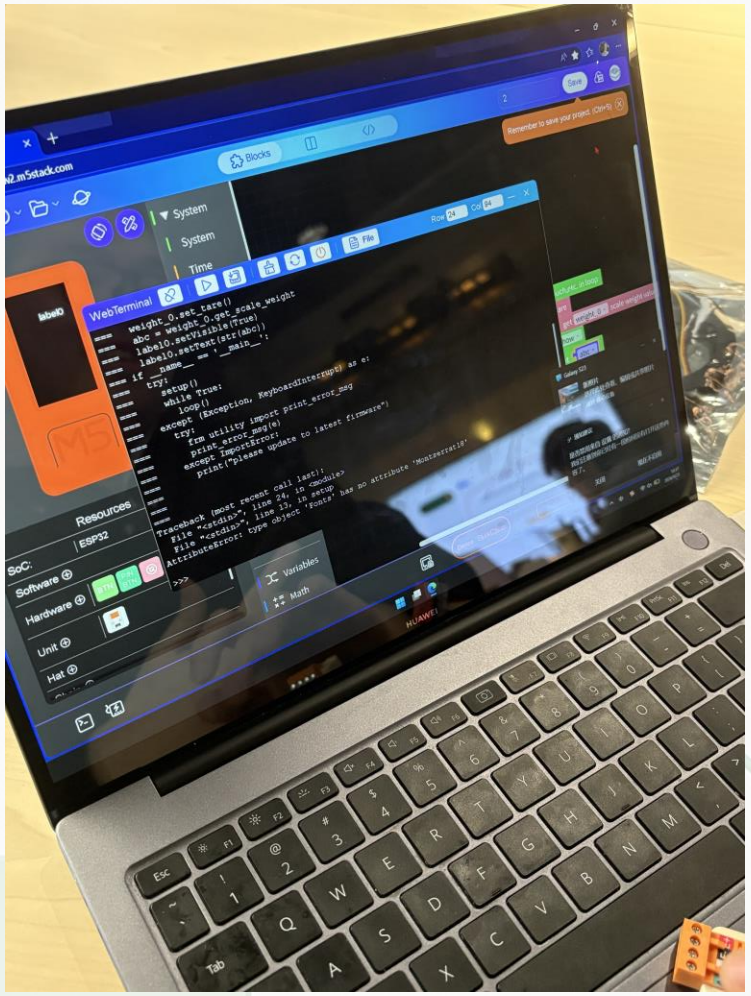
The "Add Water" prompt is unclear, leaving users confused about whether the action requires manual intervention or is automated.

Interpretation: The current instructions lack clarity regarding "manual vs. automatic" water refilling. Users need explicit guidance that clearly differentiates the two modes.

Users who travel frequently are requesting a "Temporary Caregiver" support feature.

Interpretation: This addresses a newly identified, high-frequency scenario in daily life. When the device owner is unavailable to provide care, the system should support authorizing friends or family to temporarily view device status and perform basic management tasks.

Hardware Progress & Next Steps



Completed Work

Completed 3D modeling of the food bowl; ready for the printing phase.

Weighing module functionality is basically implemented.

Individual sensor functions can work properly when tested separately.

Current Challenges

The hardware Bluetooth connection cannot yet be linked to the software app.

The six hardware components cannot currently run all functions at the same time.

The weighing module still needs further calibration to improve stability and accuracy.

Next Steps

Completed full system hardware-software integration and debugging.

Optimized algorithms to enhance the stability and accuracy of data measurement.

Conducted final system functional testing and acceptance.

Backup / Q&A

THANK YOU FOR YOUR LISTENING

GitHub

View full project source code
Technical documentation and version iteration history
https://github.com/charzzh9/PetLink.git

App

Screenshots of core features and interaction flows
Intuitive demonstration of a smooth user experience
Preview of cross-platform adaptation

Hardware

Photos of the internal circuit board and casing
Showcasing industrial design details
Structural integrity and craftsmanship

Feedback

In-depth interview recordings and test logs
Authentic user voices and feedback
Data-driven product improvement